Aditya Menon

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Profile

Biotechnology postgraduate specializing in stem cell biology, bioinformatics, and toxicology, with hands-on experience in WGS analysis and in vitro developmental models.

Education

M.Tech
SRM Institute of Science and Technology
08/2024 – Present | Chennai, India
CGPA 8.04/10.0

B.E, Biotechnology
NMAM Institute of Technology
2019 – 2023 | Nitte, India
CGPA 8.88/10.0

Work Experience

Central Research Laboratory
Junior Research Fellow
09/2023 – 03/2024 | Bengaluru, India

- Analyzed whole genome sequencing (WGS) data of Staphylococcus aureus using genome analysis tools to identify antimicrobial resistance (AMR) patterns and virulence-associated genes.
- Investigated antimicrobial resistance (AMR) profiles using genome-based approaches.
- Studied the role of PVL-carrying prophages in enhancing bacterial virulence.

Samahitha Research Solutions
Research Intern as Clinical Research Coordinator
07/2022 – 08/2022 | Hyderabad, India

- Received training in clinical trial procedures and documentation.
- Completed Case Report Forms (CRFs) using source documents for a vaccine trial.
- Conducted telephonic follow-ups to monitor participant health after treatment.
- Assisted in video documentation of the informed consent process.

Languages

English (Fluent), Hindi (Fluent), Japanese (Basic)

Projects

Environmental phthalate exposure and congenital heart disease: Investigating the role of DEHP in disrupting cardiac development using an in vitro model
M.Tech Thesis Project, SRM Institute of Science and Technology
2025 – 2026

- **Cultured mESCs** and **performed cardiac differentiation assays** to assess EDC toxicity.
- Generated embryoid bodies using **hanging drop method** for *in vitro* developmental modeling.
- Evaluated cardiac differentiation using **immunofluorescence** markers to assess the impact of phthalate exposure on developmental pathways.

Synthesis of Eco-Friendly Quantum Dots and Their Application in Bioimaging
B.E Project, NMAM Institute of Technology
2022 – 2023

- **Synthesized eco-friendly quantum dots** from lignin, chitosan, honey, and nanocellulose for bioimaging.
- **Optimized fluorescence** by phosphoric acid doping, resulting in a shift from blue to green emission.
- Characterized optical properties using **UV illumination and gel documentation.**
- **Evaluated bioimaging and antimicrobial activity** in bacterial and plant models.
- Assessed **biocompatibility** using *Poecilia reticulata*

Skills

Software Skills

- Tools: Fiji, BLAST, genome analysis platforms

Technical Skills

- Cell Culture: mESC culture, embryoid body formation, differentiation assays
- Molecular Biology: PCR, gel electrophoresis, immunofluorescence
- Bioinformatics: WGS analysis, AMR profiling, genome annotation